

Memo 06: Late-Time Spatial Distribution Checking for FVCOM-MP Cases a, b, and c

FVCOM-MP Delaware Test Cases

May 2026

Purpose

Memo 05 checked the first ten simulation days after the mass-conservation correction documented in Memo 04. This memo extends the same spatial diagnostics to later model output windows centred on simulation day 30 and simulation day 90. The focus is the microplastic tracer (`mp1`) and its floc-interaction split (`mp1_agg`, `mp1_dis`) in cases `a`, `b`, and `c`.

The practical questions are:

- (1) Do cases `a` and `b` remain identical at later times?
- (2) Does corrected case `c` continue to show active aggregation and bed exchange?
- (3) Do the late-time pointwise-maximum maps show isolated numerical accumulation artifacts?
- (4) Are the aggregated plus disaggregated fields still consistent with total `mp1`?

Methodology

The existing Python analysis script was updated to process day-specific MAT files:

- Day 10: `spatial_snapshots_{a,b,c}.mat`
- Day 30: `spatial_snapshots_{a,b,c}_day30.mat`
- Day 90: `spatial_snapshots_{a,b,c}_day90.mat`

The outputs are now organized as:

- `PYTHON/output/spatial_distribution/day10/`
- `PYTHON/output/spatial_distribution/day30/`
- `PYTHON/output/spatial_distribution/day90/`

For day 30 the spatial snapshot window is simulation days 29.500–29.957 and the station time-series window is simulation days 20.000–29.957. For day 90 the spatial snapshot window is simulation days 89.500–89.957 and the station time-series window is simulation days 80.000–89.957. Each day folder also contains a reproducible `summary_stats.csv` file with per-case maxima, means, positive-node counts, split-consistency errors, and cross-case difference magnitudes.

Numerical Summary

Table 1: Selected day-mean statistics from `summary_stats.csv`. Positive bed nodes are counted from the day-mean `bot_mass` field.

Day	Case	max <code>mp1_davg</code> kg m^{-3}	max <code>mp1_agg_davg</code> kg m^{-3}	max <code>bot_mass</code> kg m^{-2}	positive bed nodes count
30	a	3.36×10^{-4}	—	0	0
30	b	3.36×10^{-4}	0	0	0
30	c	4.62×10^{-3}	1.10×10^{-3}	1.95×10^{-1}	63 008
90	a	2.81×10^{-4}	—	0	0
90	b	2.81×10^{-4}	0	0	0
90	c	1.49×10^{-3}	6.46×10^{-5}	2.21×10^{-3}	68 351

The **b-a** differences are exactly zero in the generated summary for `mp1_davg` and `bot_mass` at both day 30 and day 90. Case **b** also retains zero aggregated mass and zero bed mass in both late-time windows. In case **c**, the maximum split-consistency error $|\text{mp1_agg} + \text{mp1_dis} - \text{mp1}|$ is $4.46 \times 10^{-10} \text{ kg m}^{-3}$ at day 30 and $1.14 \times 10^{-10} \text{ kg m}^{-3}$ at day 90 for the depth-averaged fields.

Results

R1. Day-Mean Depth-Averaged mp1: Day 30

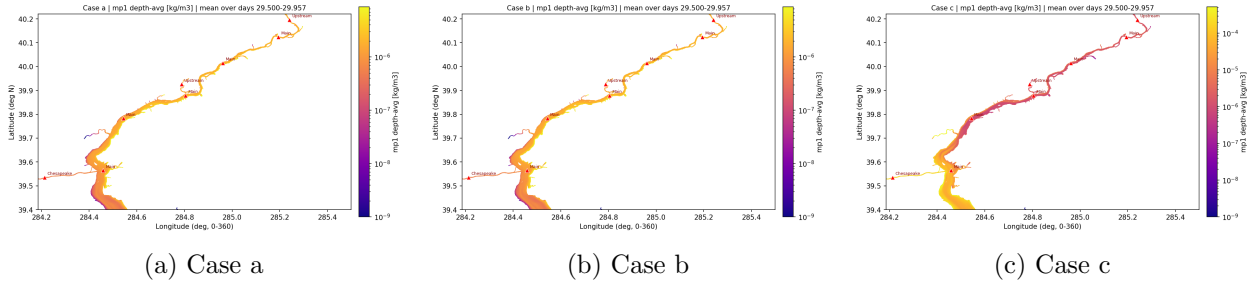


Figure 1: Day-mean depth-averaged mp1 concentration, simulation days 29.500–29.957.

R2. Day-Mean Depth-Averaged mp1: Day 90

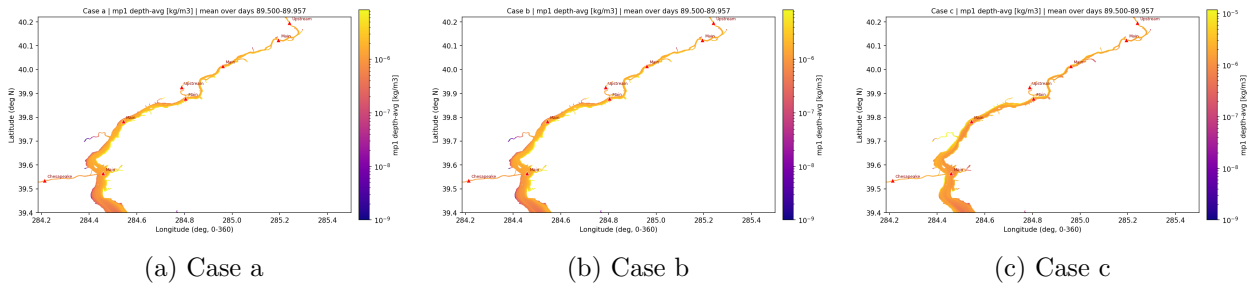


Figure 2: Day-mean depth-averaged mp1 concentration, simulation days 89.500–89.957.

R3. Floc-Component Split in Case c

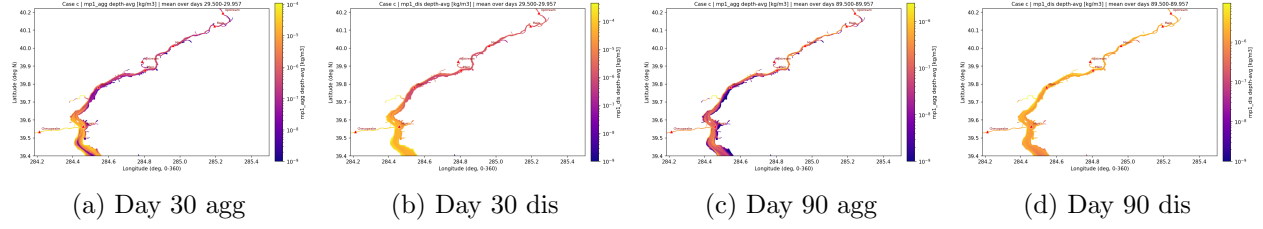


Figure 3: Depth-averaged aggregated and disaggregated plastic components in case c. Case b has zero aggregated component in both late-time windows.

R4. Bed Plastic Mass Density

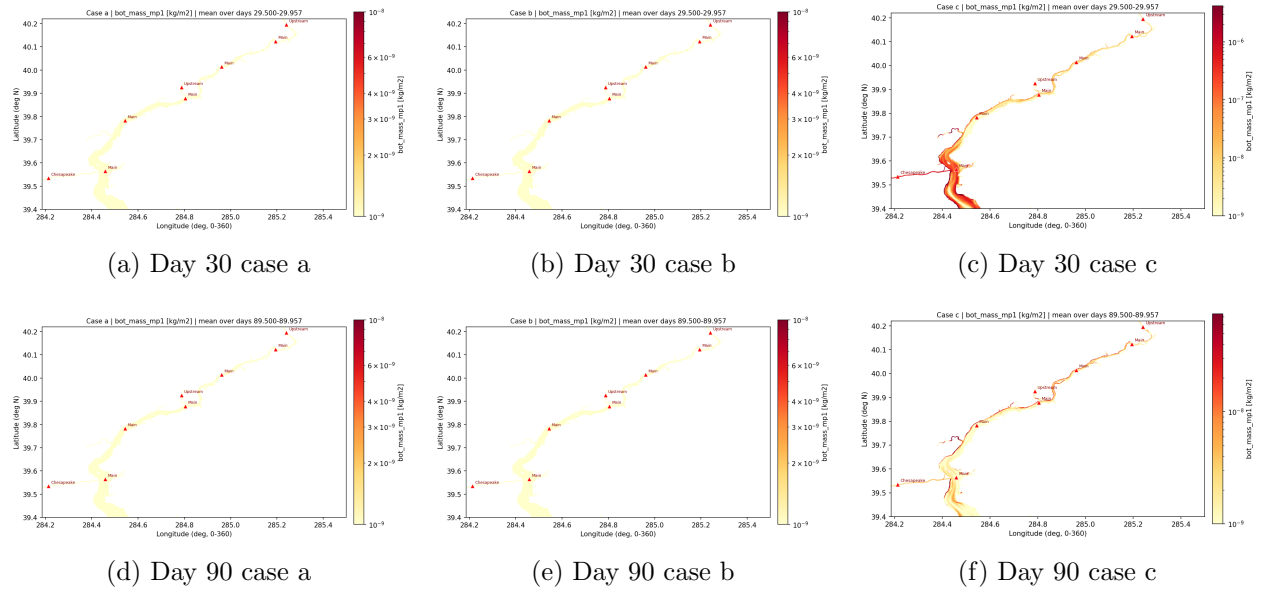


Figure 4: Day-mean bed plastic mass density for late-time windows. Cases a and b remain zero; case c retains active bed mass, with a lower point maximum in the day 90 window than in the day 30 window.

R5. Pointwise-Maximum Depth-Averaged mp1

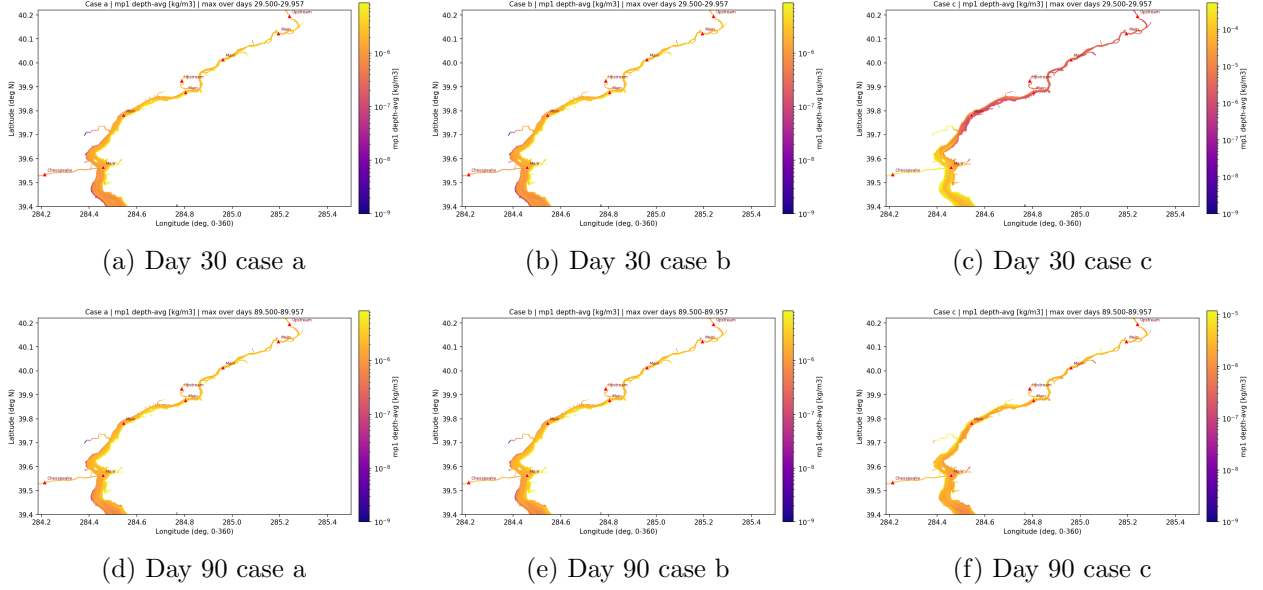


Figure 5: Pointwise maximum depth-averaged $mp1$ over each selected late-time window. The maxima remain connected to the river/bay plume structure rather than appearing as detached single-node artifacts.

R6. Cross-Case Difference Maps

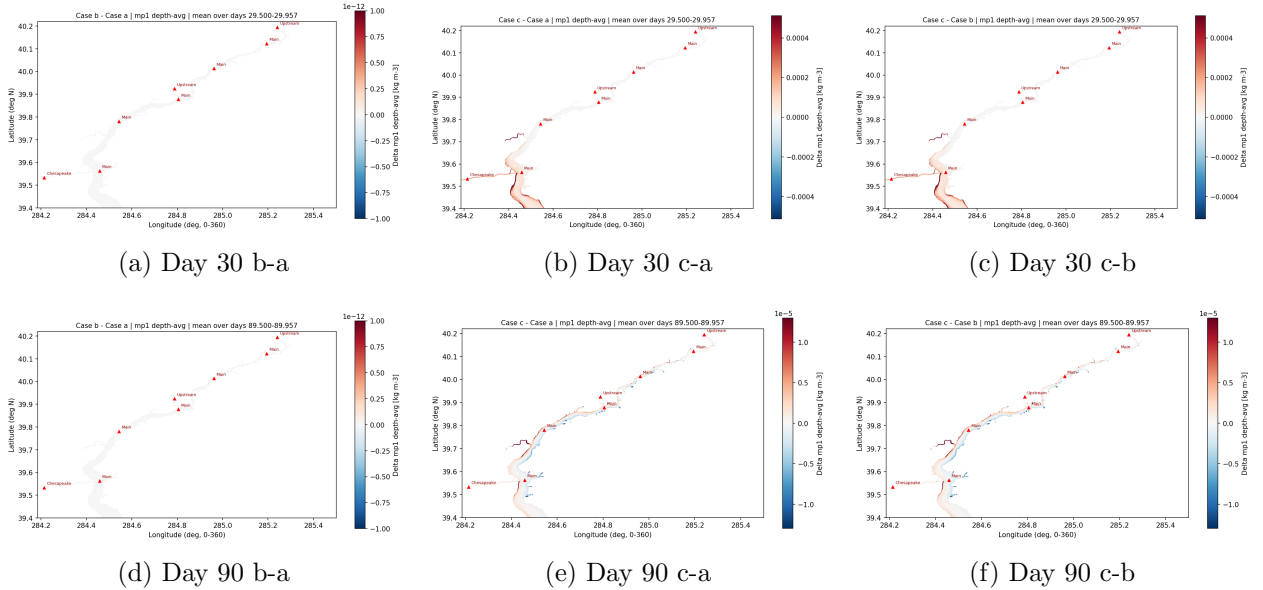


Figure 6: Day-mean cross-case differences for depth-averaged $mp1$. The b-a panels are zero in the CSV diagnostics; c-a and c-b isolate the flocculation-enabled redistribution in corrected case c.

R7. Station Time Series

Station time series - mp1 depth-averaged [kg m⁻³]



Figure 7: Depth-averaged mp1 station time series for simulation days 20.000–29.957.

Station time series - mp1 depth-averaged [kg m⁻³]

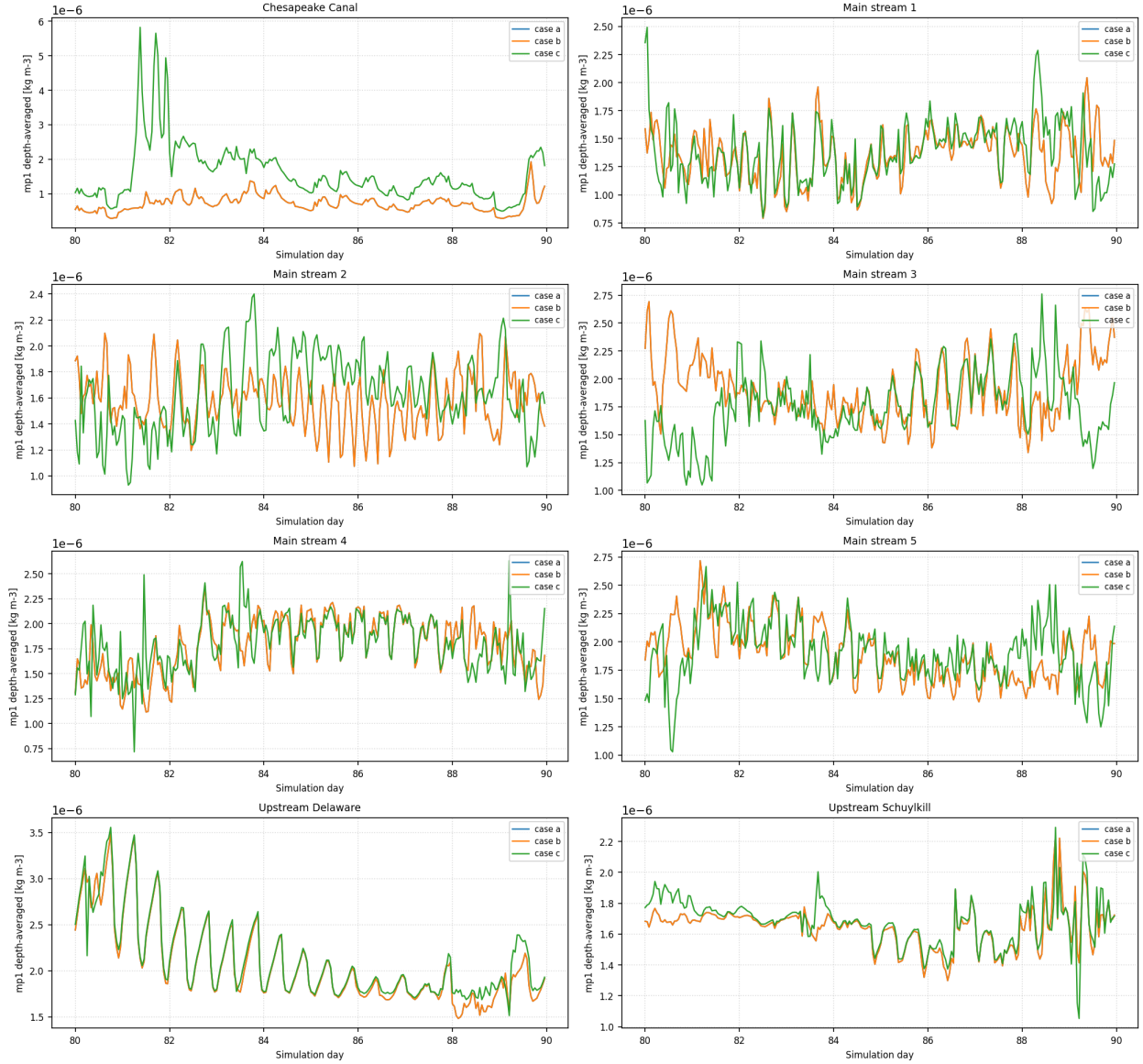


Figure 8: Depth-averaged mp1 station time series for simulation days 80.000–89.957.

Observations

- (1) **Cases a and b remain identical.** The late-time b–a differences are zero for depth-averaged mp1 and bed mass in both windows. This matches Memo 05: with the current fine-sediment parameterisation, case b does not activate measurable aggregation or bed deposition.
- (2) **Case c remains the active flocc-interaction case.** At day 30, case c has a day-mean depth-averaged mp1 maximum of $4.62 \times 10^{-3} \text{ kg m}^{-3}$ and an aggregated-component maximum of $1.10 \times 10^{-3} \text{ kg m}^{-3}$. At day 90 those maxima are lower in the selected window (1.49×10^{-3} and $6.46 \times 10^{-5} \text{ kg m}^{-3}$, respectively) but the aggregated component remains positive across nearly the full domain.

- (3) **Bed mass is present only in case c.** Cases **a** and **b** remain zero-bed-mass runs. In case **c**, the day 30 window contains a large local bed-mass maximum ($1.95 \times 10^{-1} \text{ kg m}^{-2}$), while the day 90 window has a smaller maximum ($2.21 \times 10^{-3} \text{ kg m}^{-2}$) but positive day-mean bed mass at almost all wet nodes.
- (4) **No detached numerical hot spots are evident.** The pointwise-maximum maps retain coherent plume and near-source structures. The maxima do not appear as isolated one-node artifacts disconnected from transport pathways.
- (5) **Tracer split consistency remains tight.** The maximum depth-averaged split error in case **c** remains below $5 \times 10^{-10} \text{ kg m}^{-3}$ in both late-time windows. This supports the interpretation that the late-time differences are physical/model-configuration differences rather than book-keeping drift between total, aggregated, and disaggregated tracers.

Summary

The day 30 and day 90 spatial diagnostics support the Memo 05 conclusions at later times. Case **a** and case **b** remain numerically identical under the present fine-sediment flocculation parameterisation. Corrected case **c** continues to show active aggregation, disaggregated-water-column transport, and bed exchange. The late-time fields remain spatially coherent, and the `mp1_agg+mp1_dis` consistency error remains at roundoff-level magnitude.